1 Worked example: estimability in a two-receiver network

This section makes the rank deficiency and minimum-constraint datum of the minimum-constraint datum section of the Network-mode document concrete on the smallest network that still shows the structure: two receivers $\mathcal{R} = \{A, B\}$ and four satellites $i \in \{1, 2, 3, 4\}$. We strip everything that is not essential to the estimability argument: positions are known so the geometric range is removed, a single frequency is used, the troposphere and ionosphere are absent, and the data are noise-free, so that every number below is *exact* and the gauge freedoms are visible by inspection. The point is not the size of the system but its null space.

1.1 Code only: clocks and one gauge freedom

With geometry removed, the single-frequency pseudorange of the observation-equations section collapses to a difference of two clock lumps,

$$P_r^i \;=\; a_r \;-\; g^i, \qquad a_r := c\,\mathrm{d}t_r + B_r, \quad g^i := c\,\mathrm{d}t^i + B^i, \tag{1}$$

where a_r is the receiver clock and g^i the satellite clock. On a single frequency the receiver and satellite *code biases* B_r, B^i are inseparable from the respective clocks, so each "clock" here is already a clock + code-bias lump; the IF-vs-DCB split of the satellite phase-bias section of the Network-mode document degenerates to nothing with one band. There are $|\mathcal{R}| + |\{i\}| = 2 + 4 = 6$ unknowns and $2 \times 4 = 8$ observations.

Take as ground truth (metres)

$$a_A = 5, \quad a_B = 3, \qquad g^1 = 2, \quad g^2 = -1, \quad g^3 = 4, \quad g^4 = 0.5,$$

which through Equation (1) produces the eight pseudoranges

P_r^i	$i=1$	$i=2$	$i=3$	$i=4$
$r=A$	3	6	1	4.5
$r=B$	1	4	−1	2.5

The deficiency, seen two ways. *(i) Invariance.* Add any constant κ to *all six* unknowns, $a_r \mapsto a_r + \kappa$ and $g^i \mapsto g^i + \kappa$; the difference $a_r - g^i$, and therefore every observation, is unchanged. The truth above and the shifted set $\{15, 13, 12, 9, 14, 10.5\}$ fit the data identically. Absolute clocks are unknowable; only differences are.

(ii) Singular normals. Ordering the unknowns $\boldsymbol{x} = [\,a_A, a_B, g^1, g^2, g^3, g^4\,]^{\mathrm{T}}$ and using unit weights, the normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ is

$$\boldsymbol{N} = \begin{bmatrix} 4 & 0 & -1 & -1 & -1 & -1 \\ 0 & 4 & -1 & -1 & -1 & -1 \\ -1 & -1 & 2 & 0 & 0 & 0 \\ -1 & -1 & 0 & 2 & 0 & 0 \\ -1 & -1 & 0 & 0 & 2 & 0 \\ -1 & -1 & 0 & 0 & 0 & 2 \end{bmatrix}, \qquad \boldsymbol{N}\,\mathbf{1} = \mathbf{0}, \tag{2}$$

which is 6×6 but has rank 5: the all-ones vector $\mathbf{1} = [1, 1, 1, 1, 1, 1]^{\mathrm{T}}$ is an exact null vector (every row sums to zero), and it is precisely the common-shift direction of view (i). The system $\boldsymbol{N}\boldsymbol{x} = \boldsymbol{u}$ cannot be solved without one added constraint, a *datum*.

Two datum choices, same fit. Pinning one clock removes the gauge. Two natural choices give *different* parameter values that fit the eight observations *equally and exactly*:

datum (S-basis)	a_A	a_B	g^1	g^2	g^3	g^4
$a_A \equiv 0$ (reference receiver)	0	−2	−3	−6	−1	−4.5
$g^1 \equiv 0$ (reference satellite)	3	1	0	−3	2	−1.5

Under $a_A \equiv 0$, every satellite-clock estimate comes out as $\hat{g}^i = g^i - a_A^{\text{true}}$: the reference receiver's clock is *absorbed* into all of them. The estimated "satellite clock" is not the physical clock; it is the physical clock plus the datum we chose, the *lumping* that the user must add back. This is exactly why a producer must *declare* its datum (the closed-loop section of the Network-mode document).

What is invariant. The estimable quantities (combinations annihilated by the null vector $\mathbf{1}$, i.e. differences) are identical in both bases:

estimable quantity	$a_A \equiv 0$	$g^1 \equiv 0$	(truth)
$a_A - a_B$	2	2	2
$g^2 - g^1$	−3	−3	−3
$g^3 - g^1$	2	2	2
$a_B - g^4$	2.5	2.5	$(= P_B^4)$

Only the absolute labels are convention; the differences are physics. This is the S-system of [1] in six parameters.

Remark 1 (Why the producer cannot difference). Form the double difference of this code-only model, $\nabla\Delta P_{AB}^{i,1} = (P_A^1 - P_A^i) - (P_B^1 - P_B^i)$. The between-satellite step gives $g^i - g^1$ for *both* receivers, so the between-receiver step yields exactly $\mathbf{0}$. Double differencing annihilates the entire model, because DD is built to delete clocks and here clocks are all there is (the differencing section). A DD user would estimate nothing; the network producer instead *keeps* the clocks and pins one datum. That degeneracy is the producer's reason for existence in miniature.

1.2 Adding carrier phase: the satellite phase bias needs the network

Now add the single-frequency carrier phase. With the same clock lumps it reads

$$\Phi_r^i \;=\; a_r - g^i \;+\; b^i \;+\; N_r^i, \tag{3}$$

where b^i is the satellite phase bias, the producer's target, and N_r^i is the float ambiguity (in metres). The receiver phase bias and the receiver code-minus-phase offset are constant per receiver and have been absorbed into N_r^i; we keep b^i explicit because it is what the producer must emit. Extend the truth with

$$b^1 = 0.10, \qquad b^2 = -0.20, \qquad b^3 = 0.05, \qquad b^4 = 0.15,$$
$$N_A^i : 10,\ -5,\ 20,\ 8, \qquad N_B^i : 12,\ -3,\ 18,\ 9,$$

giving the eight carrier-phase observations

Φ_r^i	$i=1$	$i=2$	$i=3$	$i=4$
$r = A$	13.10	0.80	21.05	12.65
$r = B$	13.10	0.80	17.05	11.65

Counting: one receiver vs. two. The clocks are fixed by the code, so per satellite the phase contributes $b^i + N_r^i$. With *one* receiver this is a single equation $b^i + N_A^i$ in two unknowns: the satellite phase bias has no separate existence at all: it is inseparable from that one receiver's ambiguity, and not even redundant. With *two* receivers there are two equations,

$$b^i + N_A^i, \qquad b^i + N_B^i, \tag{4}$$

in three unknowns (b^i, N_A^i, N_B^i): rank 2, one deficiency per satellite. The estimable two-dimensional space now contains the *between-receiver single-difference ambiguity* $\nabla N_{AB}^i = N_A^i - N_B^i$, a quantity that simply does not exist with one receiver, and is exactly the integer a user resolves (the differencing section). The second receiver is what gives b^i an identity shared across stations.

Datum and solution. The joint code + phase system has a 1-dimensional clock gauge (as before) plus one phase-bias↔ambiguity gauge per satellite: $b^i \mapsto b^i + c_i$, $N_r^i \mapsto N_r^i - c_i$ for all r. The minimum-constraint datum used here is the clock pin $a_A \equiv 0$, S-transform-equivalent to the reference-*satellite* clock pin $g^1 \equiv 0$ that the estimator's S-basis actually declares (the minimum-constraint datum section of the Network-mode document: the shared-block gauge pins a satellite clock per constellation, not a receiver clock), plus the reference-receiver ambiguity pins $N_A^i \equiv 0$ for every i (the local datum on the pivot receiver, folded into station A's rows). Solving gives

	$i=1$	$i=2$	$i=3$	$i=4$
emitted phase bias $\bar{b}^i = b^i + N_A^i$	10.10	−5.20	20.05	8.15
$\hat{N}_A^i$ (datum)	0	0	0	0
$\hat{N}_B^i = N_B^i - N_A^i$	2	2	−2	1

(the clocks are unchanged from Section 1.1). One checks the fit directly, e.g. $\hat{\Phi}_B^3 = a_B - g^3 + \bar{b}^3 + \hat{N}_B^3 = -2 - (-1) + 20.05 + (-2) = 17.05$, matching the observation.

Reading the result. Two things are worth dwelling on.

- **The emitted bias is dominated by the lump.** The physical satellite phase bias of satellite 1 is $b^1 = 0.10$ m, but the *emitted* estimable bias is $\bar{b}^1 = 10.10$ m: it has absorbed receiver A's ambiguity (10 m) for that satellite. This is Odijk's $\bar{b}^i = b^i + (\text{pivot-receiver ambiguity})$ (the satellite phase-bias section of the Network-mode document, Eq. 4/9) made arithmetic. A consumer recovers integers only by removing the *same* lumped quantity: producer datum and consumer integer recovery must be one S-system.
- **The network unlocks the integer.** The estimable $\hat{N}_B^i = N_B^i - N_A^i = \nabla N_{AB}^i$ is the between-receiver single-difference ambiguity, the very quantity an RTK user fixes to an integer. It is identically absent in a single-receiver problem; the second station is what brings it into existence. The satellite phase bias becomes a definite, transferable number for the price of pinning one reference ambiguity, and that pin is what it carries.

This four-satellite, two-receiver instance is, up to the geometry and the second frequency it suppresses, exactly the synthetic network the estimator core is validated on: the `RecoversSyntheticTruthUpToSTransform` test (`tests/test_network_estimator.cpp`) builds rows of exactly this shape (at four receivers and five satellites, from a random truth), pins the reference-satellite clock datum $g^1 \equiv 0$ (S-transform-equivalent to the $a_A \equiv 0$ used above) through `NetworkEstimator::solve`, and asserts recovery of the satellite-side parameters *up to the declared S-transform*, i.e. recovery of the lumped $\bar{b}^i$ and the single-difference ambiguities above, not of the unobservable physical b^i.

References

[1] P. J. G. Teunissen. "Generalized inverses, adjustment, the datum problem and S-transformations". In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.